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School of Management

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Professor
Vrins Frédéric

Assistant
Vanderveken Rodolphe

Authors
Basile Houriez - 39502100
Nicolas Poncin - 76152100

Group 2
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Declaration on the Use of Artificial Intelligence

Artificial Intelligence (ChatGPT, OpenAI) was used solely as an assistive tool for improving the Python implementation and enhancing the quality, clarity and readability of the written report. All mathematical developments, theoretical analyses, modelling choices, proposed extensions, implementation ideas and financial reasoning were entirely developed by the authors, who take full responsibility for the correctness, originality and accuracy of all results presented herein.

The following market parameters are assumed :

$$S_0 = 30, \quad K = 30, \quad \sigma = 0.15, \quad r = 0.04, \quad T = 1$$

Part I

Absolute & Quadratic Payoffs

I.1 Payoffs Construction Using Vanilla Options

Absolute Payoff

We consider the absolute payoff at maturity T,

$$\psi_T^a = |S_T - K|.$$

By definition of the absolute value, this payoff can be written piecewise as

$$|S_T - K| = \begin{cases} S_T - K, & \text{if } S_T \geq K, \\ K - S_T, & \text{if } S_T < K. \end{cases}$$

Using standard option payoff notation, this decomposition becomes

$$\psi_T^a = (S_T - K)^+ + (K - S_T)^+.$$

This expression shows that the absolute payoff is obtained by combining a European Call & European Put at the same Strike K. This is renowned as a straddle.

The resulting payoff is symmetric around the strike K and grows linearly with the distance of S_T from K, as illustrated in Figure 1.

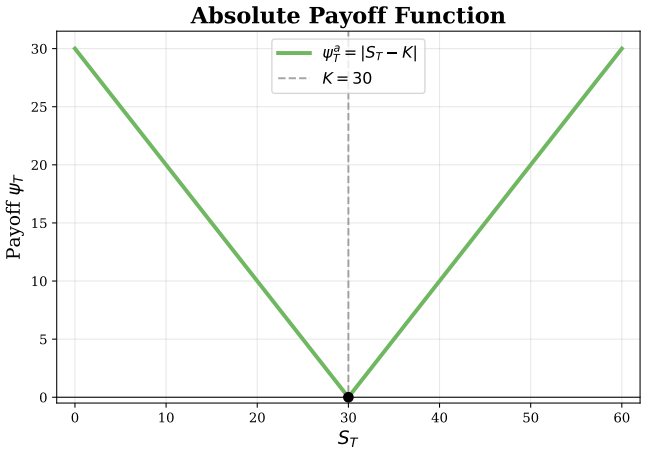


Figure 1

Quadratic Payoff

The quadratic payoff at maturity T is defined as

$$\psi_T^q = (S_T - K)^2.$$

This convex payoff cannot be replicated with a single call and put, as vanilla options have only piecewise linear structures. A portfolio of options across various strikes would be necessary to match the quadratic shape.

The resulting payoff is quadratic in the distance from K, as shown in Figure 2.

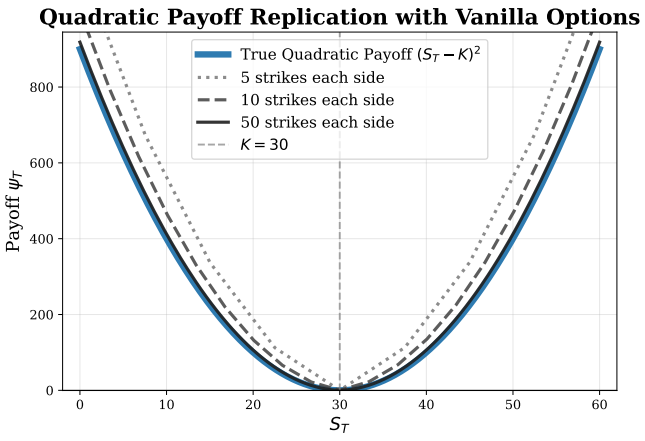


Figure 2

*For each side, the strikes are placed symmetrically around K for n contracts.

I.2 Comparison Absolute & Quadratic

The Absolute payoff grows linearly with $|S_T - K|$, while the Quadratic payoff grows exponentially with $(S_T - K)^2$. For $|S_T - K| < 1$, the Absolute payoff dominates. Beyond this threshold and thanks to its convexity, the Quadratic payoff accelerates and becomes larger.

With $S_0 = 30$, **a price move exceeding ± 1 (i.e., $\pm 3.33\%$)** is required for the Quadratic to outperform. This threshold effect should be carefully considered when advising the client.

Furthermore, the directional bias of the stock is a critical factor. If the client has a strong **upward bias**, the Quadratic payoff becomes increasingly advantageous. Due to its convexity, the spread between the two payoffs widens as prices rise, making the Quadratic product more profitable in bullish market conditions.

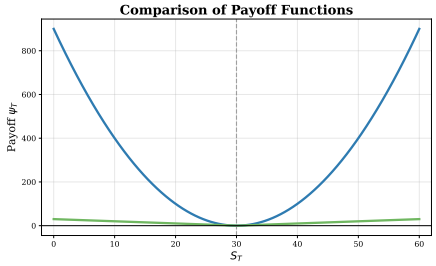


Figure 3

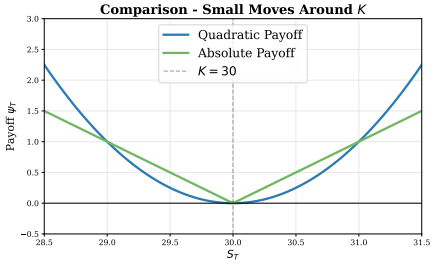


Figure 4

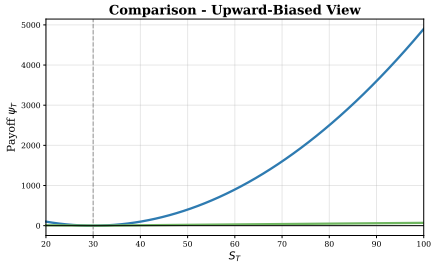


Figure 5

I.3 Intuitive Pricing

Intuitively, V_0^q is expected to be higher than V_0^a if the market assigns enough probability to large moves, because the quadratic payoff rewards tails much more strongly. If volatility is low or extreme moves are unlikely, then V_0^a can be higher.

By the Law of One Price and since the Quadratic payoff dominates the Absolute payoff whenever $|S_T - K| > 1$, its price is higher in expectation thanks to its convexity.

$$\text{Price}(\psi_T^q) > \text{Price}(\psi_T^a),$$

The **Quadratic payoff** rewards:

- **High volatility environments and large deviations**
- **Long maturity product**
- **Heavy-tailed beliefs about S_T**

The reverse holds only in the narrow band $|S_T - K| < 1$, then the **Absolute payoff** may be more valuable, which occurs in:

- **Extremely low volatility ($\sigma \to 0$) environments, where deviation is unlikely**
- **Very short time to maturity ($T \to 0$)**

Part II

Pricing

In this section, we price the two contracts introduced previously. Under the risk-neutral measure, the underlying asset price follows a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_t^\star, \tag{1}$$

where $W^\star$ denotes a Brownian motion under the risk-neutral measure. The closed-form solution (see Appendix A for the complete derivation) is:

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right), \quad Z \sim \mathcal{N}(0,1). \tag{2}$$

II.1 Monte Carlo Pricing

To price a product using Monte Carlo simulation, we generate N independent paths of the underlying asset following a GBM and compute the payoff $\psi_T^{(i)}$ for each path. The price is then estimated as:

$$V_0^{MC} = e^{-rT} \cdot \frac{1}{N}\sum_{i=1}^{N} \psi_T^{(i)}.$$

By the Law of Large Numbers, convergence is achieved as $N \to \infty$. Figure 6 shows GBM paths (top-left) with drift $r = 0.04$ and volatility $\sigma = 0.15$, terminal distribution (top-right), and payoff convergence (bottom). The Absolute payoff converges faster; the Quadratic requires more paths due to its convexity.

As $N \to \infty$, the Monte Carlo estimator converges to the true price. As shown in Figure 6 (bottom-right) $N = 10{,}000$ paths are sufficient to achieve an accurate estimate for the Absolute payoff, while the Quadratic payoff requires significantly more paths due to its higher variance.

To ensure convergence of the Monte Carlo estimator, we simulate $N = 100{,}000$ paths. The resulting price estimates are:

$$V_0^{MC,a} = 3.6395, \qquad V_0^{MC,q} = 22.7422.$$

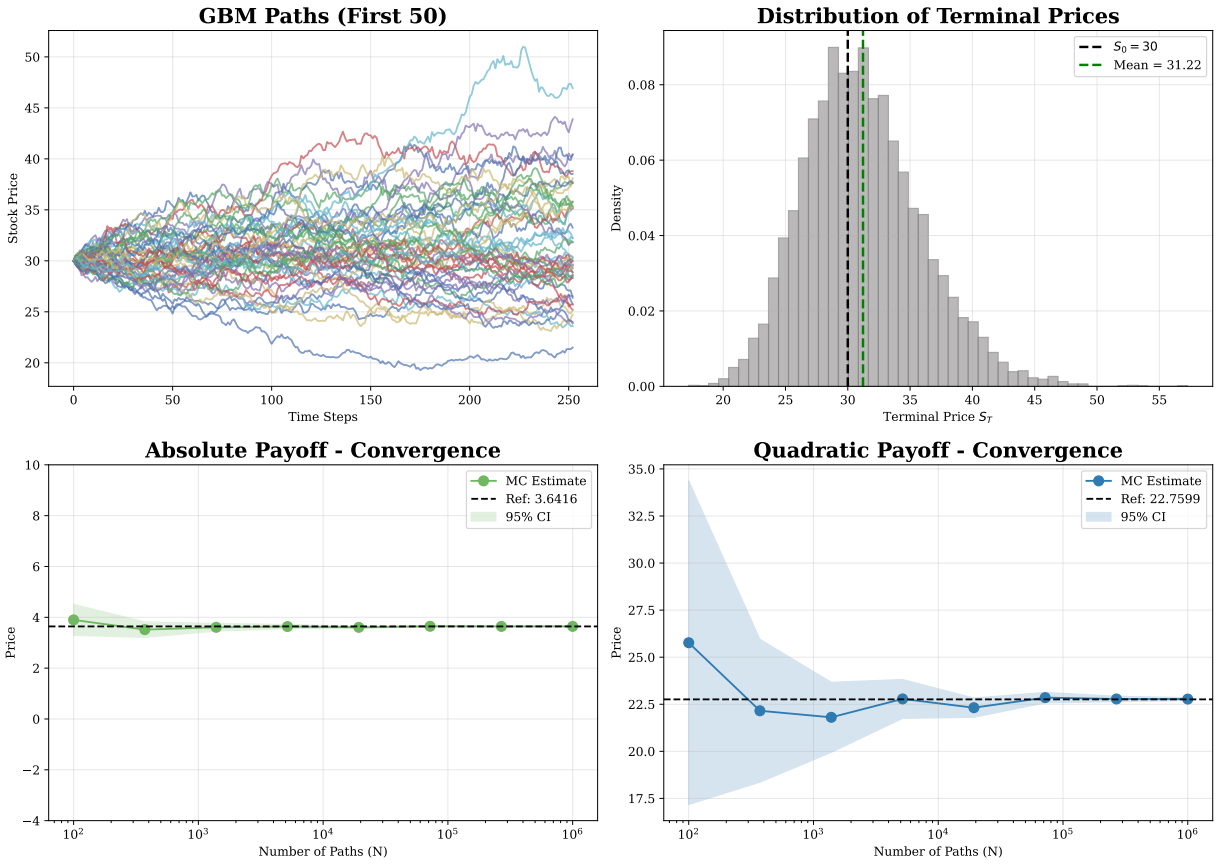


Figure 6

II.2 Numerical Integration

For a continuous random variable X with density $p_X(x)$, its expectation is given by

$$\mathbb{E}[\psi(X)] = \int_{-\infty}^{\infty} \psi(x)\, p_X(x)\, dx.$$

As shown in Appendix B, under the risk-neutral measure $\mathbb{P}^*$, the terminal asset price S_T follows a lognormal distribution. Consequently, the time-zero value of any European payoff $\psi(S_T)$ is obtained as the discounted expectation

$$V_0 = e^{-rT}\mathbb{E}^*[\psi(S_T)] = e^{-rT}\int_0^{\infty} \psi(s)\, p_{S_T}(s)\, ds,$$

where the lognormal density of S_T is

$$p_{S_T}(s) = \frac{1}{s\sqrt{2\pi\sigma^2 T}} \exp\left(-\frac{\left(\ln s - \ln S_0 - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right), \qquad s > 0.$$

Numerical Implementation

The numerical integration consists in evaluating

$$I = \int_0^{\infty} \psi(s)\, p_{S_T}(s)\, ds, \qquad \Rightarrow \qquad V_0 = e^{-rT} I.$$

For visualization purposes, the integral is approximated on a uniform grid using the composite **trapezoidal rule**. Let

$$g(s) = \psi(s) p_{S_T}(s),$$

then

$$\int_a^b g(s)\, ds \approx \frac{h}{2}\left[g(s_0) + 2\sum_{i=1}^{N-1} g(s_i) + g(s_N)\right], \qquad h = \frac{b-a}{N}, \qquad s_i = a + ih.$$

To analyse the contribution of different terminal prices, the integration is performed over the truncated domain $[0.01, 100]$ using a uniform grid of $N = 100\,000$ points.

Table 1: Numerical Integration Results

Option	Integral I	Price V_0	Error
Absolute $\psi_T^a = \lvert S_T - K\rvert$	3.7894	3.6408	7.44×10^{-9}
Quadratic $\psi_T^q = (S_T - K)^2$	23.6841	22.7555	1.49×10^{-8}

Figure 7 illustrates the numerical integration process. The left panel shows the risk-neutral lognormal density of S_T, the middle panel displays the integrands $\psi(s) f_{S_T}(s)$, whose areas correspond to the expected payoffs, and the right panel presents their cumulative contributions to the option prices. The results show that the quadratic payoff derives a larger fraction of its value from high terminal asset prices than the absolute payoff, reflecting its greater sensitivity to the right tail of the distribution due to its stronger convexity.

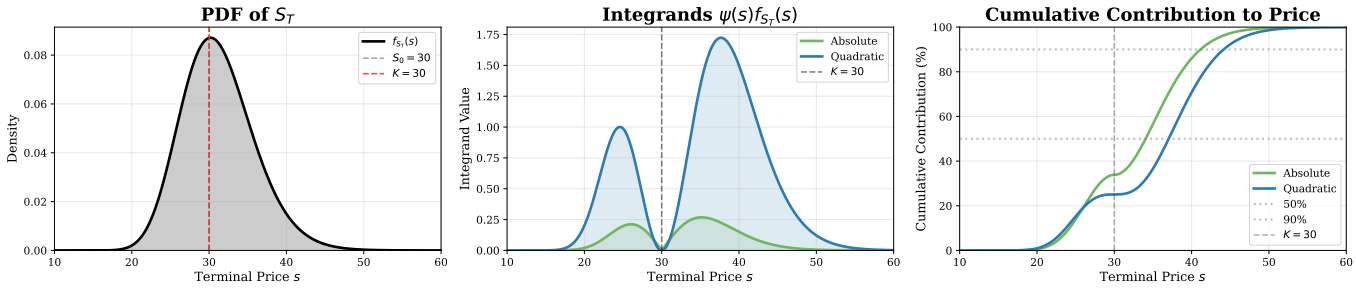


Figure 7

II.3 Analytical Integration

The idea behind analytical pricing is to derive closed-form solutions for the two pricing formulas.

Quadratic Payoff

The quadratic payoff can be expanded as:

$$(S_T - K)^2 = S_T^2 - 2KS_T + K^2.$$

Using the risk-neutral pricing formula, the time-0 price is:

$$V_0^q = e^{-rT}\mathbb{E}^*[(S_T - K)^2] = e^{-rT}\left(\mathbb{E}^*[S_T^2] - 2K\mathbb{E}^*[S_T] + K^2\right).$$

To evaluate this expression, we need the first and second moments of S_T under the risk-neutral measure.

First Moment: The Martingale Property

Under the risk-neutral measure $\mathbb{P}^*$, the discounted asset price $e^{-rt}S_t$ is a martingale. This means that for any $t \leq T$:

$$e^{-rt}S_t = \mathbb{E}^*[e^{-rT}S_T \mid \mathcal{F}_t].$$

Setting $t = 0$, we obtain:

$$S_0 = e^{-rT}\mathbb{E}^*[S_T] \quad \Longrightarrow \quad \mathbb{E}^*[S_T] = S_0 e^{rT}.$$

Second Moment: Lognormal Moment Formula

The martingale property only gives us the first moment. For the second moment $\mathbb{E}^*[S_T^2]$, we must use the lognormal distribution of S_T.

From the GBM solution: $S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right), \quad Z \sim \mathcal{N}(0,1).$

For a lognormal random variable $X = e^Y$ with $Y \sim \mathcal{N}(\mu, \sigma^2)$, the second moment is:

$$\mathbb{E}[X^2] = \exp\left(2\mu + 2\sigma^2\right).$$

Here, $\mu = \ln S_0 + (r - \frac{1}{2}\sigma^2)T$ and $\sigma^2 = \sigma^2 T$. Therefore:

$$\mathbb{E}^*[S_T^2] = \exp\left(2\ln S_0 + 2\left(r - \frac{1}{2}\sigma^2\right)T + 2\sigma^2 T\right).$$

Simplifying:

$$\mathbb{E}^*[S_T^2] = S_0^2 \exp\left(2rT - \sigma^2 T + 2\sigma^2 T\right) = S_0^2 e^{(2r+\sigma^2)T}.$$

Yielding the analytical price of the quadratic contract:

$$V_0^q = e^{-rT}\left(S_0^2 e^{(2r+\sigma^2)T} - 2KS_0 e^{rT} + K^2\right) \approx 22.7555.$$

Absolute Payoff

As shown in Section I.I, in the risk-neutral valuation, the absolute payoff decomposes as

$$V_0^a = e^{-rT}\mathbb{E}^*[|S_T - K|] = e^{-rT}\mathbb{E}^*[(S_T - K)^+] + e^{-rT}\mathbb{E}^*[(K - S_T)^+] = C_0(K,T) + P_0(K,T).$$

Pricing the European Call

By definition, the call price can be written as:

$$C_0(K,T) = e^{-rT}\mathbb{E}^*[(S_T - K)^+] = e^{-rT}\left(\mathbb{E}^*[S_T \mathbf{1}_{\{S_T>K\}}] - K\mathbb{P}^*(S_T > K)\right).$$

We first determine the probability that the option finishes in the money. The condition $S_T > K$ can be rewritten in terms of the standard normal variable Z as:

$$S_T > K \iff Z > \frac{\ln(K/S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} =: -d_2.$$

Since $Z \sim \mathcal{N}(0,1)$:

$$\mathbb{P}^*(S_T > K) = \mathbb{P}^*(Z > -d_2) = \Phi(d_2), \qquad d_2 = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

For the truncated first moment, by the martingale property $\mathbb{E}^*[S_T] = S_0 e^{rT}$, we can use a change of measure to obtain:

$$\mathbb{E}^*[S_T \mathbf{1}_{\{S_T>K\}}] = S_0 e^{rT}\Phi(d_1),$$

where:

$$d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

Substituting back yields the Black-Scholes price of a European call option:

$$C_0(K,T) = e^{-rT}\left(S_0 e^{rT}\Phi(d_1) - K\Phi(d_2)\right) = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2).$$

Pricing the European Put

Rather than recomputing the expectation explicitly, we rely on the put-call parity relation:

$$C_0(K,T) - P_0(K,T) = S_0 - Ke^{-rT},$$

which directly yields:

$$P_0(K,T) = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1).$$

Adding the call and put prices and using the identity $\Phi(-x) = 1 - \Phi(x)$, we obtain:

$$V_0^a = S_0\left(2\Phi(d_1) - 1\right) - Ke^{-rT}\left(2\Phi(d_2) - 1\right).$$

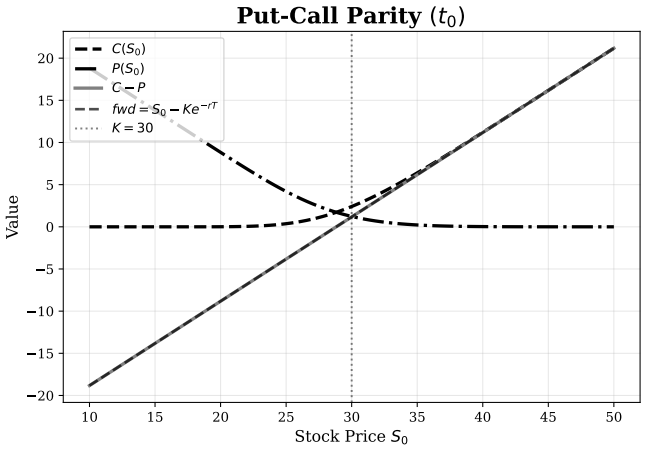


Figure 8

With the project parameters, we get $d_1 \approx 0.342$, $d_2 \approx 0.192$, $\Phi(d_1) \approx 0.6337$, $\Phi(d_2) \approx 0.5760$, and thus $V_0^a \approx 3.64$, which matches the Monte Carlo and numerical integration results.

Price Comparison

Table 2: Price Comparison

Method	Absolute	Quadratic
Monte Carlo	3.6395	22.7422
Numerical Integration	3.6408	22.7555
Analytical pricing	3.6394	22.7555

The three methods are in excellent agreement, validating our pricing implementations. The minor Monte Carlo discrepancy for the quadratic payoff is due to sampling error. The quadratic contract is significantly more expensive than the absolute contract due to its convexity, which amplifies large price movements.

II.4 Impact of Key Parameters on Option Price

As seen in the Analytical Pricing section, the closed-form formulas derived using the Black-Scholes framework allow us to analyze how each parameter affects the option price:

$$V_0^a = S_0\left(2\Phi(d_1) - 1\right) - Ke^{-rT}\left(2\Phi(d_2) - 1\right), \qquad V_0^q = e^{-rT}\left(S_0^2 e^{(2r+\sigma^2)T} - 2KS_0 e^{rT} + K^2\right),$$

$$\text{where } d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \text{ and } d_2 = d_1 - \sigma\sqrt{T}.$$

1. **Variation in S_0**

   The option price increases monotonically with the initial underlying price S_0 for both payoff structures. The quadratic payoff exhibits a steeper slope due to its convexity, while the absolute payoff shows a more linear relationship. An interesting observation is that since the stock price is bounded below by zero, the downside profit is capped, making both products cheaper for downward moves. This is illustrated by the leftward shift in Figure 9.

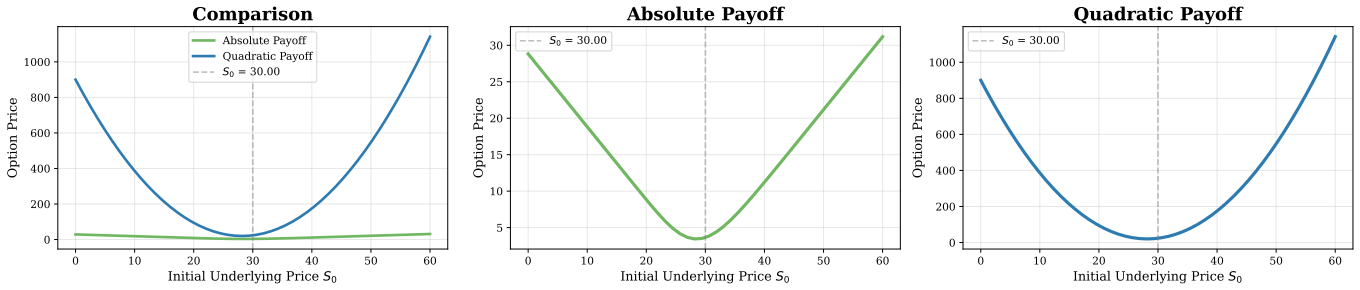


Figure 9: Option price as a function of S_0

2. **Variation in T**

   As time to maturity T increases, both option prices rise due to greater uncertainty and time value. The absolute payoff increases at a decreasing rate, exhibiting a concave behavior with respect to maturity, while the quadratic payoff grows at an increasing rate, displaying convex behavior due to its stronger sensitivity to variance. Notably, for short-term maturities, the absolute payoff can be more expensive than the quadratic payoff.

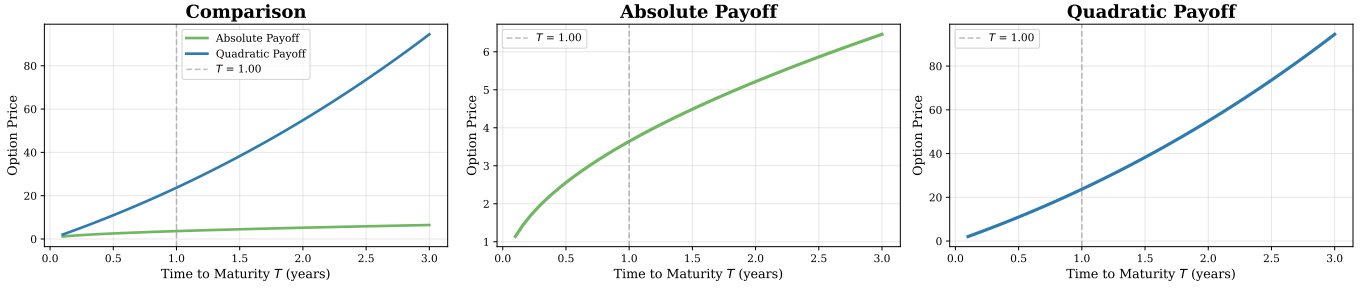


Figure 10: Option price as a function of time to maturity T

3. **Variation in Strike (K)**

   The option price exhibits a convex relationship with the strike price K. The quadratic payoff reaches its minimum at $K = S_0 e^{rT}$, while the absolute payoff displays a V-shaped pattern with its minimum at the same point.

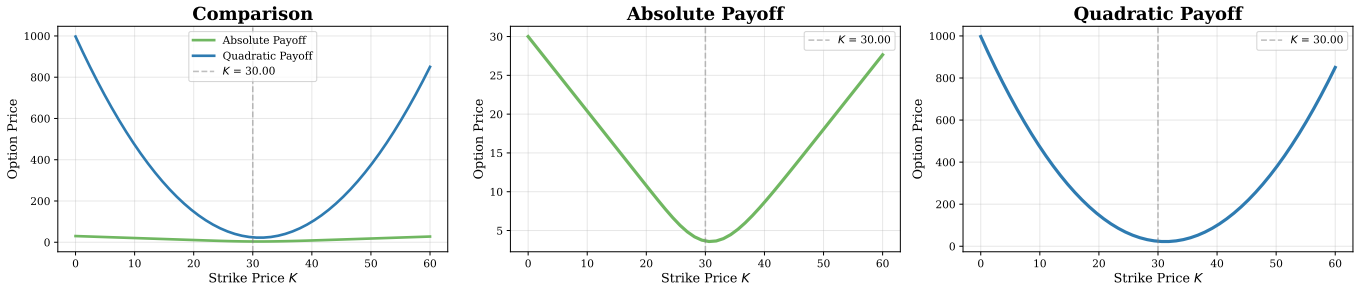


Figure 11: Option price as a function of strike K

4. **Variation in Volatility (σ)**

   Both option prices increase with volatility σ, reflecting the positive convexity of the payoffs. However, the quadratic payoff shows significantly stronger sensitivity (quadratic growth) compared to the absolute payoff, which increases at a slower, more linear rate.

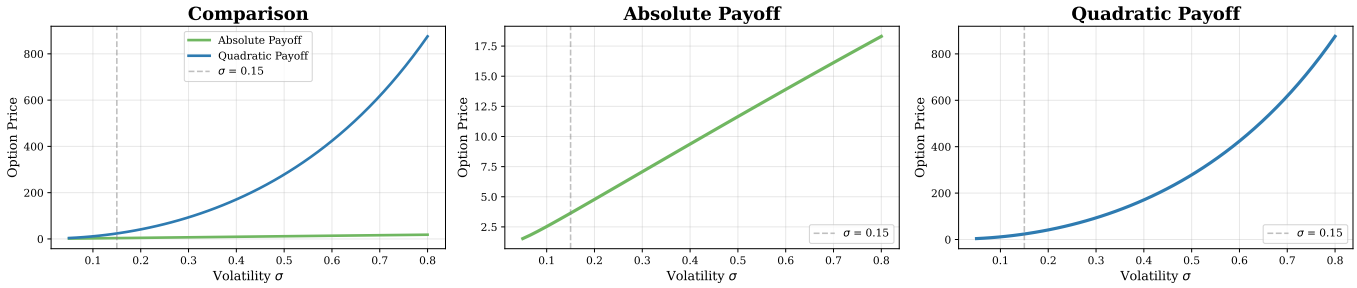


Figure 12: Option price as a function of volatility σ

II.5 Options Delta

The delta,measures the sensitivity of the option price to small changes in the initial stock price and is a key risk metric for hedging purposes. It defined as the first derivative with respect to the underlying price at time t, is:

$$\Delta_t = \frac{\partial V_t}{\partial S_t}.$$

Quadratic Payoff

For a time interval $[t, T]$, the quadratic payoff price is:

$$V_t^q = e^{-r(T-t)} \left(S_t^2 e^{(2r+\sigma^2)(T-t)} - 2KS_t e^{r(T-t)} + K^2 \right).$$

Taking the derivative:

$$\Delta_t^q = e^{-r(T-t)} \left(2S_t e^{(2r+\sigma^2)(T-t)} - 2Ke^{r(T-t)} \right).$$

Simplifying:

$$\Delta_t^q = 2S_t e^{(r+\sigma^2)(T-t)} - 2K.$$

At inception ($t = 0$), this reduces to:

$$\Delta_0^q = 2S_0 e^{(r+\sigma^2)T} - 2K \approx 3.869.$$

Absolute Payoff

As stated in II.3, The absolute payoff: $V_t^a = C_t + P_t$. Its delta is: $\Delta_t^a = \Delta_t^{\text{call}} + \Delta_t^{\text{put}}$.

Under Black-Scholes, with $\tau = T - t$:

$$\Delta_t^{\text{call}} = \Phi(d_1(t)), \qquad \Delta_t^{\text{put}} = \Phi(d_1(t)) - 1.$$

Therefore:

$$\Delta_t^a = 2\Phi(d_1(t)) - 1.$$

At inception ($t = 0$):

$$\Delta_0^a = 2\Phi(d_1) - 1, \quad \text{with} \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

Using the project parameters, $d_1 \approx 0.35$, giving $\Phi(d_1) \approx 0.6368$, and thus: $\Delta_0^a \approx 0.267$.

Delta as a Function of Different Parameters

The tables in Appendix-C illustrate the behavior of the delta for both payoffs under changes in parameters such as $(S_0, K, \sigma, T, t, r)$. Here are the main conclusions.

The sensitivity analysis reported in Tables 5–9 highlights the distinct behaviour of the two payoffs. The delta of the absolute payoff is bounded between -1 and 1 and exhibits a smooth transition around the strike, although the transition is not perfectly symmetric in S_0 due to the lognormal dynamics and the positive risk-free rate. In contrast, the quadratic payoff produces an unbounded delta that varies almost linearly with S_0 and K, while increasing much more rapidly with volatility because it depends on the second moment of the terminal stock price. Both deltas increase with maturity and the risk-free rate, and decrease as the remaining time to maturity shrinks.

As shown in Figure 13, the delta of the absolute payoff initially decreases with volatility because the term r/σ dominates in d_1, before reaching a minimum around $\sigma \approx \sqrt{2r}$ and increasing again as the $\sigma/2$ term becomes dominant.

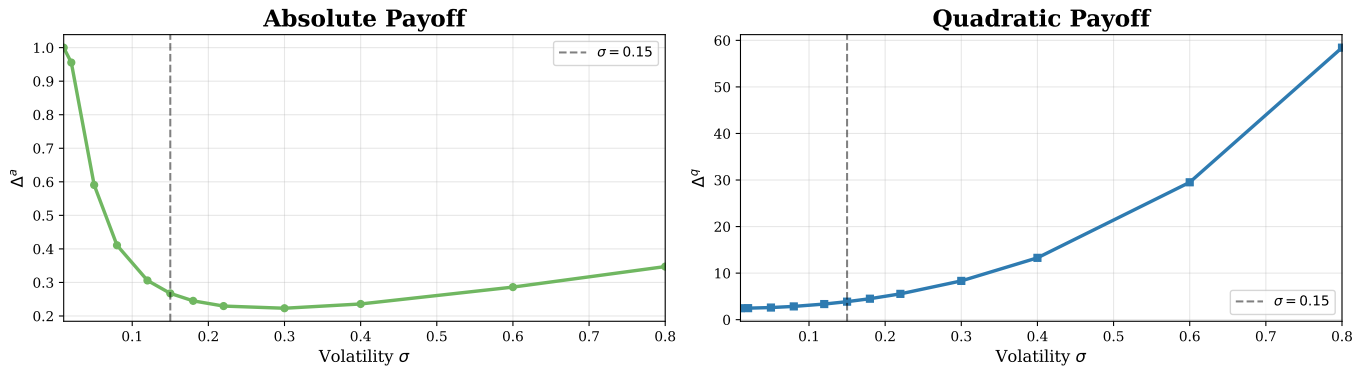


Figure 13: Sensitivity Analysis of Delta with respect to Volatility (σ)

II.6 Comparison between products

(i) Payoff Function ψ_T

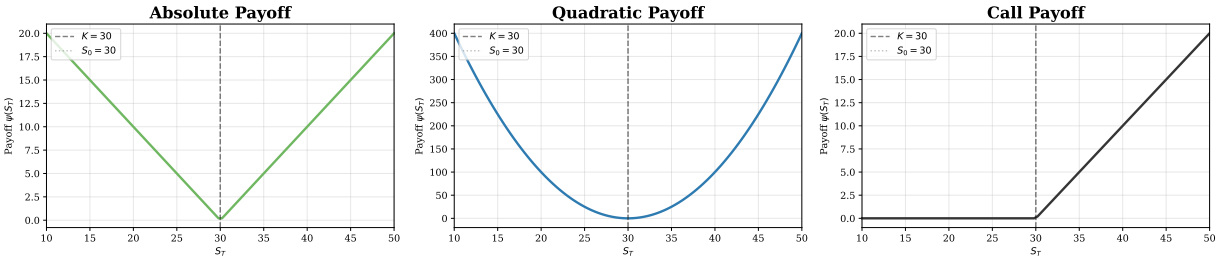


Figure 14

The three payoff structures exhibit fundamentally different shapes as functions of S_T. The Absolute payoff is V-shaped and symmetric around K, growing linearly in either direction. The Quadratic payoff is U-shaped and strictly convex, growing much faster than the Absolute payoff away from K. The Call payoff is asymmetric: it is flat (zero) for $S_T \leq K$ and grows linearly only above K, reflecting its purely directional nature.

(ii) Price Function V_t

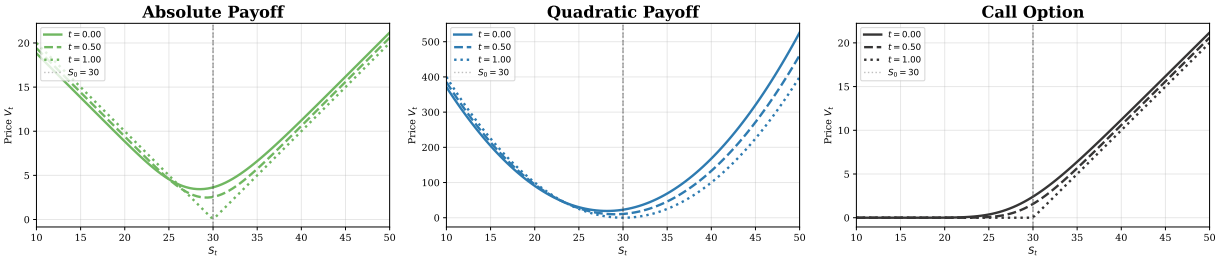


Figure 15

Comparing V_t at $t \in \{0, T/2, T\}$, the Quadratic option is consistently the most expensive across all three dates, as the standard deviation of S_T is large enough that $x^2 \gg x$ in expectation.

The Call is the cheapest because it captures one side of the distribution (upside only, payoff is zero if $S_T \leq K$).The Absolute payoff is equivalent to a straddle (Call + Put), so it captures both sides. Mechanically, $V_0^a = C_0 + P_0 > C_0$.

(iii) Profit Function $\psi_T - V_0$

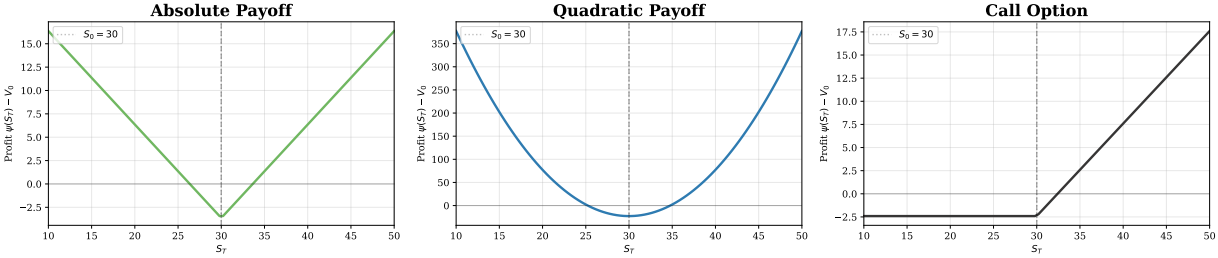


Figure 16

The profit profiles inherit the shape of the payoffs, net of the initial premium. The Absolute payoff's profit remains symmetric around K, shifted downward by V_0^a. The Call's profit is asymmetric, with a single break-even point at $K + V_0^{call}$, capped downside loss (limited to the premium paid), and unlimited upside.

(iv) Delta Function Δ_t

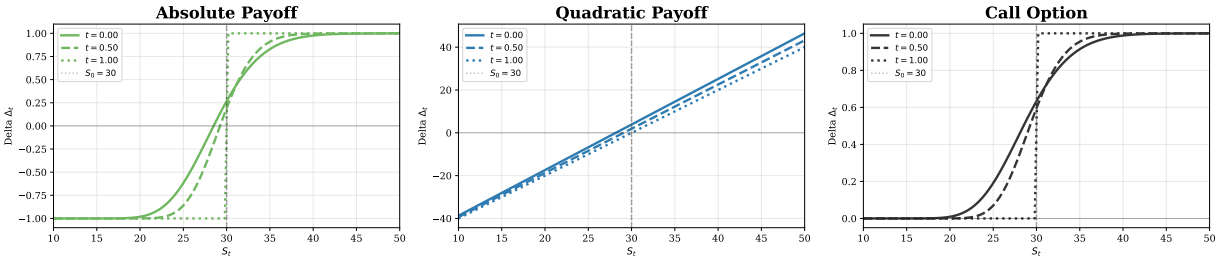


Figure 17

The Delta profiles reveal distinct hedging behaviors. The Absolute payoff's Delta is bounded in $[-1, 1]$, reflecting its piecewise-linear, symmetric payoff. The Quadratic payoff's Delta is unbounded and linear in S_t, a direct consequence of its convexity (constant positive Gamma). The Call's Delta is bounded in $[0, 1]$, consistent with its one-sided exposure. For all three products, as $t \to T$, the Delta curves converge toward the slope of the terminal payoff function.

Part III

Replication and Hedging

III.1 Replication Strategy - No transaction Cost

To replicate the payoffs of the absolute and quadratic contracts, we employ a **dynamic delta hedging strategy**. The objective is to construct a self-financing portfolio consisting of the underlying asset and a risk-free bank account that replicates the option's payoff at maturity (figure-19):

$$\Pi_T = \Delta_{T-1} S_T + \beta_T = \Psi_T = V_T$$

By the *Law of One Price*, the initial value of this replicating portfolio must equal the option price, ensuring no arbitrage (developed in Appendix-B).

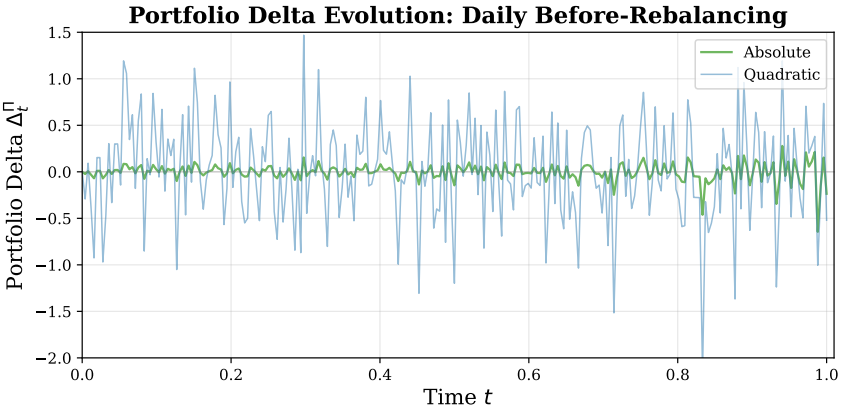


Figure 18

The key idea in theory is to continuously adjust the number of shares held in the underlying asset such that the portfolio remains **delta-neutral** at all times. While continuous re-balancing is not possible since it is a discrete action, the trader chooses a discrete dates $t_i = i\delta$ time to rebalance. The delta at each step is recomputed dynamically using the current spot S_{t_i} and the remaining time to maturity $\tau_i = T - t_i$. In the example shown in figure-18, rebalancing occurs daily. The delta of the option, as stated in II.3 is, $\Delta_t = \partial V_t / \partial S_t$. By holding Δ_t shares, we offset the option's directional exposure, making the portfolio insensitive to small movements in the underlying. The hedging error at maturity is defined as the difference between the option payoff and the final portfolio value: $\varepsilon_T = \psi(S_T) - \Pi_T$.

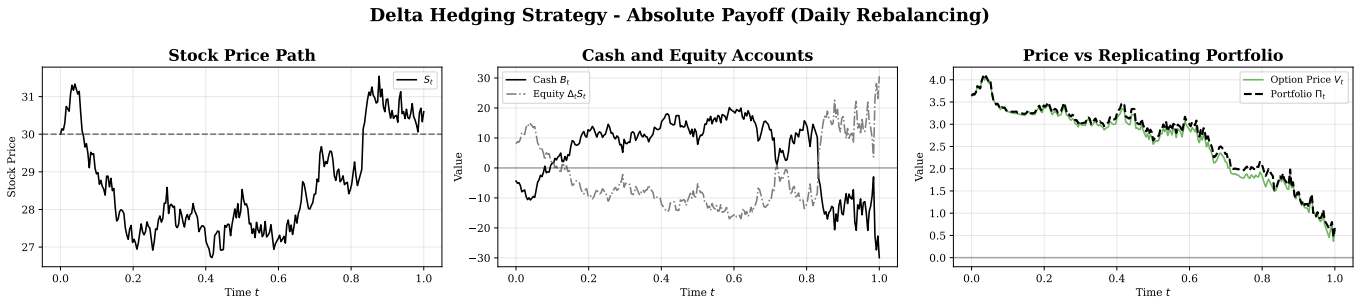


Figure 19: Delta Hedging Strategy for Absolute Payoff: Stock Price Path (left), Cash and Equity Accounts (center), Option Price vs Replicating Portfolio (right).

The same delta hedging strategy applies to both the absolute and quadratic payoffs. The only difference lies in the delta formula used for each contract.

To assess the impact of discrete re-balancing on hedging performance, we simulate the delta hedging strategy for both payoffs using four different re-balancing frequencies: daily, weekly, monthly, and quarterly.

Table 3: Hedging Error Statistics (No Transaction Costs)

	Absolute Payoff			**Quadratic Payoff**		
Interval	Mean	Std. Dev.	RMSE	Mean	Std. Dev.	RMSE
Daily	0.0000	0.1964	0.1936	−0.2164	6.7937	6.8864
Weekly	−0.0128	0.4352	0.4386	−0.2799	7.9224	7.9338
Monthly	−0.0323	0.8296	0.8498	0.3669	11.4060	11.5378
Quarterly	0.0060	1.4695	1.4423	−0.6243	16.9053	17.5368

The main observations are:

- For the absolute payoff, Δ_t^a converges to ± 1 as $\tau \to 0$, depending on whether S_T ends above or below K. While Δ_t^q converges to $2(S_T - K)$ as $\tau \to 0$.
- Additionally, the quadratic payoff exhibits a larger tracking error than the absolute payoff at the same rebalancing frequency due to its higher convexity (gamma), making it more sensitive to discrete adjustments.
- Increasing rebalancing frequency (smaller δ, e.g. daily vs monthly) reduces the tracking error for both payoffs, confirming convergence toward the continuous-time replication result. (see Figure 20)

At first sight, the quadratic payoff appears more difficult and more costly to hedge, due to its higher gamma exposure. This convexity requires a more frequent and larger rebalancing of the hedging portfolio, which intuitively raises concerns about the inclusion of transaction costs.

III.2 Replication Strategy - With Transaction Costs

When transaction costs of $c = 50$ bps per dollar traded are applied, the hedging errors become significantly biased and increase in magnitude. The quadratic payoff incurs substantially larger costs because its delta changes more rapidly, requiring more frequent and larger rebalancing trades.

At each rebalancing date, the cost incurred is $c \cdot |\Delta_{t_{i+1}} - \Delta_{t_i}| \cdot S_{t_{i+1}}$, deducted directly from the cash account; the initial trade at $t = 0$ (buying Δ_0 shares) also incurs a transaction cost, which reduces the initial cash balance B_0.

	Absolute Payoff			**Quadratic Payoff**		
Interval	Mean	Std. Dev.	RMSE	Mean	Std. Dev.	RMSE
Daily	1.4940	0.6452	1.6274	19.3704	4.1103	19.8017
Weekly	0.6675	0.5333	0.8544	8.1513	7.2462	10.9065
Monthly	0.3517	0.8586	0.9278	3.8949	11.6885	12.3203
Quarterly	0.2853	1.4847	1.5119	1.8724	17.6030	17.7023

Table 4: Hedging Error Statistics (With Transaction Costs $c = 50$ bps)

The absolute payoff exhibits much smaller hedging errors across all intervals (0.2853 to 1.4940) compared to the quadratic payoff (1.8724 to 19.3704).

Unlike section III.1, **the strategy is no longer perfectly self-financing** net of frictions: costs represent a real drag on portfolio value that accumulates over the rebalancing dates, this reflects the cumulative cost of hedging, which introduces a trade-off between rebalancing frequency and transaction costs: more frequent rebalancing reduces the discretization error but increases the cumulative costs. This is particularly visible in the daily set up of figure 20

In figure 21 and 22 the absolute payoff (green) maintains errors close to zero throughout the life of the option, with only minor deviations. The quadratic payoff (blue) exhibits significantly larger errors, particularly when transaction costs are included (dashed lines).

The replication performance of both payoffs is illustrated in Figure 23, both payoffs track the option value reasonably well without transaction costs. However, when transaction costs are introduced, the quadratic payoff's portfolio value diverges more significantly from the option price. The absolute payoff remains relatively stable, confirming that it is more suitable for replication purposes.

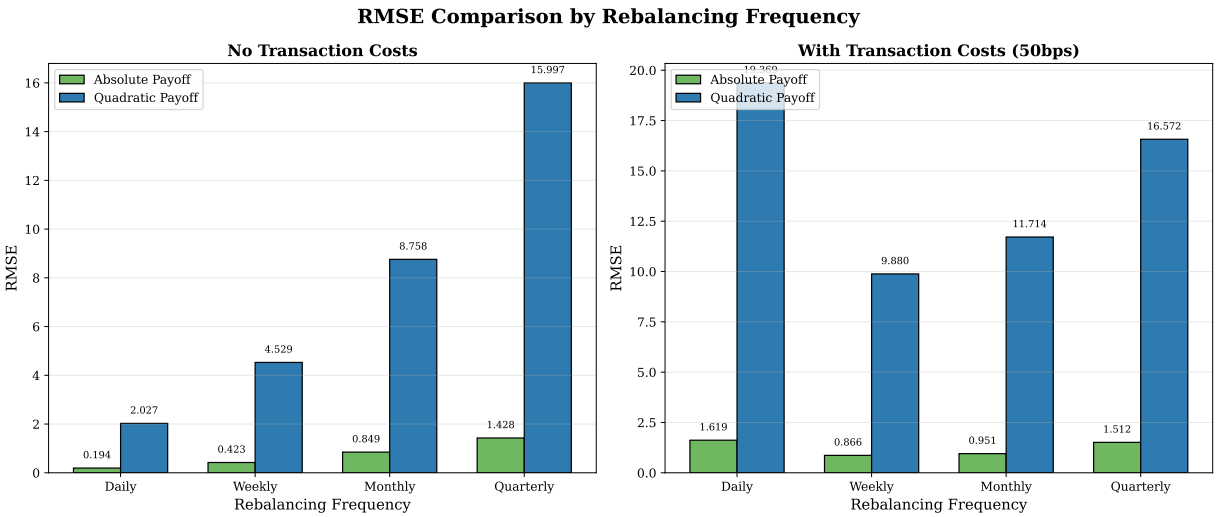


Figure 20: RMSE Comparison

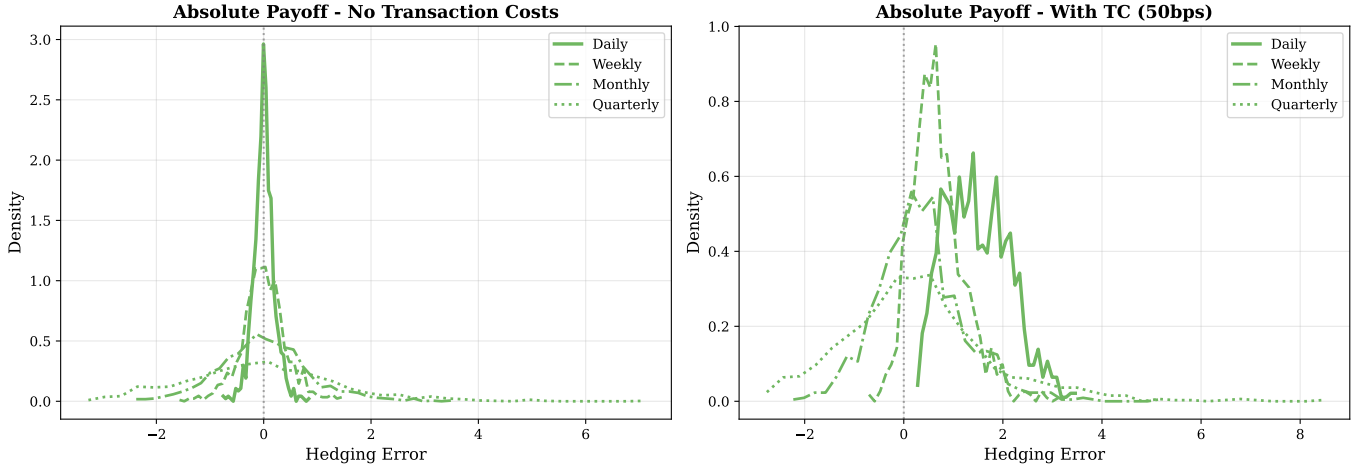


Figure 21: Quadratic Payoff distribution comparison

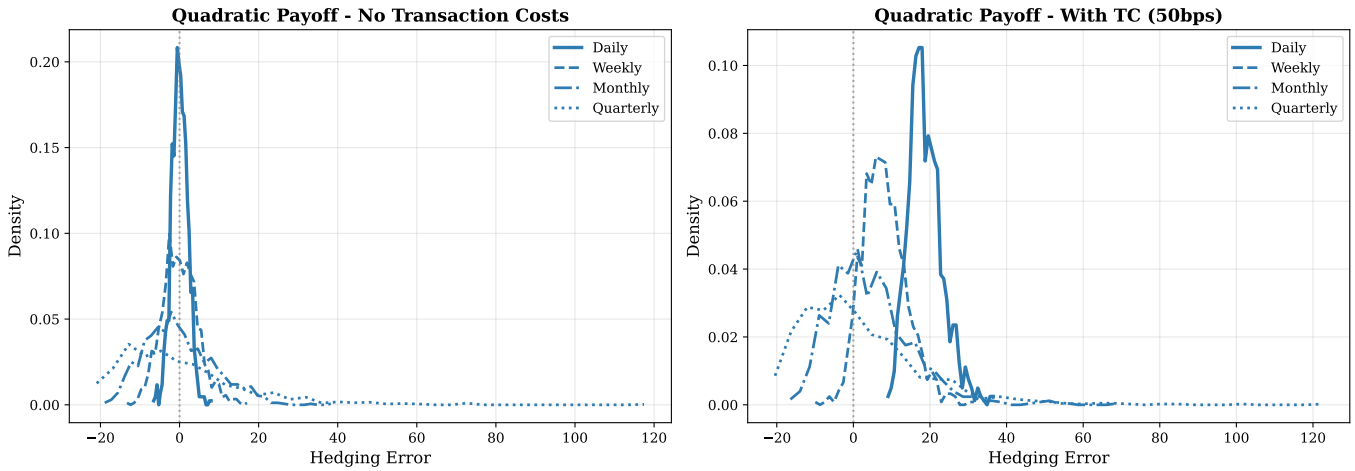


Figure 22: Quadratic Payoff distribution comparison

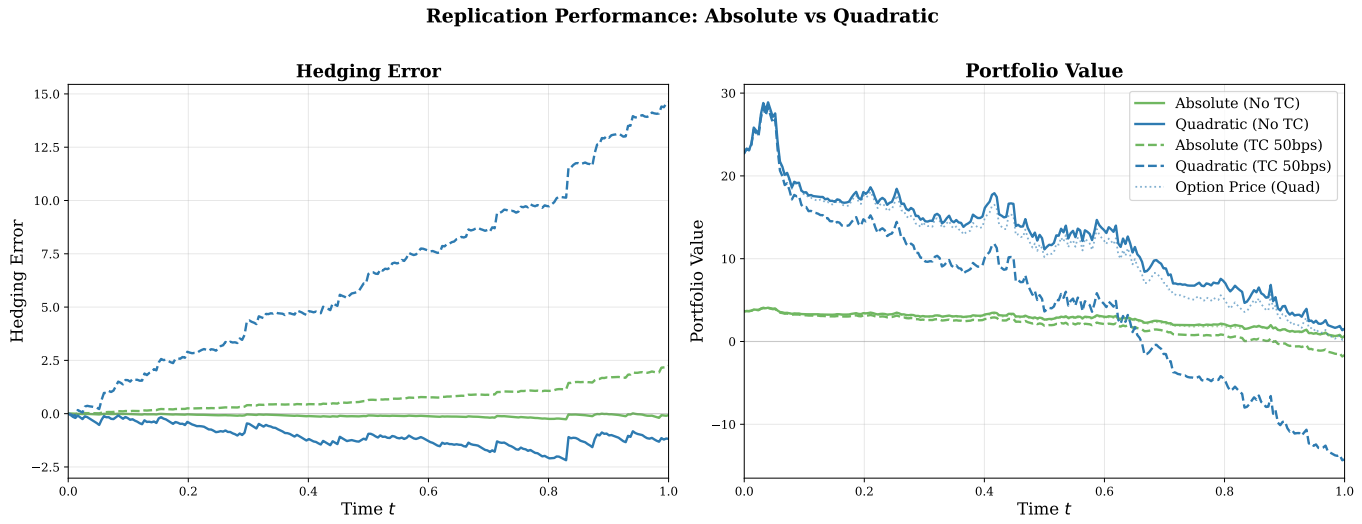


Figure 23

III.3 Recommendation for the Client

Since the client expects heavy variations in the price of S, intuition would suggest offering the **Quadratic payoff**, as it delivers a strictly better payoff than the Absolute payoff whenever $|S_T - K| > 1$, rewarding large moves far more strongly through its convexity. However, it requires more frequent portfolio adjustments and incurs considerably higher transaction costs, which must be absorbed to generate a net benefit.

To quantify this trade-off, we simulated a hedging portfolio with an initial capital of \$1,000,000 for each product, estimating the expected cumulative transaction costs and the maximum number of contracts that could be issued while remaining fully hedged. The methodology is described in appendix D and the results, summarized in Table 11, show that although the Quadratic payoff commands a much higher theoretical premium upfront ($V_0^q = 22.76$ vs $V_0^a = 3.64$), this premium is largely eroded in practice: its higher gamma forces more frequent and larger rebalancing, so that transaction costs consume 46.61% of the collected premium, compared with only 4.78% for the Absolute payoff. For the same \$1,000,000 portfolio, this translates into a net collected premium of \$682,090 for the Quadratic payoff against \$954,367 for the Absolute payoff, despite the latter issuing a much larger number of contracts (262,128 vs 29,975) at a lower price.

For the seller, the larger premium does not compensate for the additional hedging risk and cost it entails: large, unhedged moves between rebalancing dates are more costly precisely when the discrete hedge is most likely to fail, and the underlying would need to exhibit far more extreme and sustained variation than assumed here for the Quadratic payoff to remain preferable once hedging frictions are included. Even after introducing Delta-Gamma hedging and adaptive rebalancing in section III.4, the product still remains expensive to replicate.

Conversely, the **Absolute payoff** is significantly easier to hedge: its lower gamma results in a smaller tracking error at every rebalancing frequency, and fewer, smaller rebalancing trades translate into lower cumulative transaction costs. This lower residual hedging error also means less exposure to unhedged P&L swings, and the strategy is operationally simpler, with fewer trades to execute and monitor and therefore lower execution risk, which allows for more competitive pricing to the client.

We recommend issuing the **Absolute payoff**, as it offers the best compromise between client exposure, replication accuracy, operational execution, and implementation cost.

III.4 Extensions and improvements

Extension 1: Improved Market Model for Delta–Gamma Hedging

The main limitation of a pure delta hedging strategy is that it only eliminates the first-order sensitivity of the payoff to the underlying asset. For highly convex products, such as the quadratic payoff, the remaining gamma exposure becomes the primary source of replication error, particularly between rebalancing dates. To improve the hedging performance while preserving the completeness of the Black–Scholes market and satisfying both FTAP I and FTAP II, we enlarge the hedging portfolio by introducing a liquid European call option.

The hedging portfolio therefore becomes

$$\Pi_t = \alpha_t S_t + \beta_t B_t + \eta_t C_t,$$

where S_t denotes the underlying stock, B_t the risk-free asset and C_t a traded European call written on the same underlying. The hedge is constructed by matching both the gamma and the delta of the product,

$$\eta_t = \frac{\Gamma_t^P}{\Gamma_t^C}, \qquad \alpha_t = \Delta_t^P - \eta_t \Delta_t^C,$$

where Γ_t^P and Δ_t^P denote the gamma and delta of the target product, while Γ_t^C and Δ_t^C denote the corresponding Greeks of the traded European call used for hedging.

while the remaining wealth is invested in the risk-free asset. Since the call also carries vega exposure, the resulting portfolio is less sensitive to changes in volatility. Although the Black–Scholes market remains complete, the enlarged hedging portfolio provides a more accurate replication of highly convex payoffs.

Results: The numerical results show that this additional hedging instrument mainly benefits products with significant gamma exposure.

For the quadratic payoff, the Delta–Gamma strategy reduces the monthly RMSE from 11.11 to 7.93, corresponding to an improvement of approximately 29%, while the average transaction cost increases only moderately from 4.30 to 5.71. In contrast, the Absolute payoff can be written exactly as a static combination of the underlying stock, a European call and the risk-free asset. Consequently, the Delta–Gamma portfolio provides an exact static replication of the product and the remaining RMSE of 0.181 is entirely explained by the initial transaction cost, independently of the rebalancing frequency. These results confirm that extending the market model is particularly effective for highly convex derivatives, whereas products admitting a static replication require virtually no dynamic hedging.(Full results in Table 12 in Appendix E).

Extension 2: Adaptive Rebalancing Strategy

Another important practical limitation is the presence of transaction costs. As shown in Section 2.3, increasing the rebalancing frequency reduces the replication error but substantially increases the cumulative trading costs. This trade-off is particularly pronounced for the quadratic payoff, whose large gamma causes the hedge ratio to fluctuate rapidly, requiring frequent portfolio adjustments.

Rather than rebalancing at fixed dates, we propose an event-driven strategy. Let

$$E_t = \Pi_t - V_t$$

denote the instantaneous hedging error, where Π_t is the value of the replicating portfolio and V_t is the theoretical value of the product. The portfolio is monitored daily and rebalanced only when

$$|E_t - \mu_E| > k\sigma_E, \qquad k \in \{1, 2, 3\},$$

where μ_E and σ_E are respectively the mean and standard deviation of the daily hedging-error distribution obtained from the fixed-frequency strategy. This adaptive rule aims to reduce unnecessary transactions while maintaining an acceptable replication accuracy by intervening only when the hedging error becomes statistically significant.

Results: This approach directly targets the trade-off between replication accuracy and transaction costs. For the quadratic payoff under Delta hedging, increasing the threshold from 1σ to 3σ reduces the average number of rebalancing operations from 226 to 160, while transaction costs decrease from 17.02 to 12.36. Although the hedging error becomes slightly more dispersed, the RMSE remains relatively stable, decreasing from 18.45 to 17.48. The improvement is even more significant under Delta–Gamma hedging, where increasing the threshold reduces the average number of trades from 189 to only 7, lowering the average transaction cost from 13.20 to 2.99. Despite a moderate increase in RMSE for the largest threshold, the adaptive strategy provides a substantially better compromise between replication accuracy and implementation costs than classical calendar-based rebalancing. (Full results in Table 12 in Appendix E).

Limitations: The proposed extensions remain within the Black–Scholes framework and therefore assume constant volatility, continuous market liquidity and frictionless trading apart from proportional transaction costs. In practice, stochastic volatility, jumps, liquidity constraints and imperfect option markets would further affect the replication performance and should be incorporated in future work.

Part IV

Appendix

A GBM solution

Under the risk-neutral measure, the underlying asset price follows a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_t^\star$$

where $W^\star$ denotes a Brownian motion under the risk-neutral measure.

To obtain the transition density, we apply Itô's lemma to the log-process $X_t = \ln S_t$:

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2}(dS_t)^2.$$

Substituting $dS_t = rS_t dt + \sigma S_t dW_t^\star$ and noting that $(dS_t)^2 = \sigma^2 S_t^2 dt$:

$$d(\ln S_t) = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t^\star.$$

Integrating from 0 to T:

$$\ln S_T - \ln S_0 = \left(r - \frac{1}{2}\sigma^2\right) T + \sigma(W_T^\star - W_0^\star).$$

Since $W_T^\star - W_0^\star \sim \mathcal{N}(0, T)$, we can write $W_T^\star - W_0^\star = \sqrt{T}Z$ with $Z \sim \mathcal{N}(0, 1)$. Thus:

$$\ln S_T = \ln S_0 + \left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}Z.$$

Exponentiating both sides yields the lognormal transition:

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T}Z\right), \quad Z \sim \mathcal{N}(0, 1).$$

B Lognormal Distribution of S_T

We prove that under the risk-neutral measure, the terminal stock price S_T follows a lognormal distribution.

The SDE for the Stock Price

Under the risk-neutral measure $\mathbb{P}^*$, the stock price S_t follows a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_t^\star, \tag{B.1}$$

where $W_t^\star$ is a Brownian motion under $\mathbb{P}^*$.

Applying Itô's Lemma

Let $X_t = \ln S_t$. Applying Itô's lemma to $f(S_t) = \ln S_t$:

$$d(\ln S_t) = \frac{1}{S_t} dS_t - \frac{1}{2S_t^2}(dS_t)^2. \tag{B.2}$$

Substituting (B.1) and using $(dS_t)^2 = \sigma^2 S_t^2 dt$:

$$d(\ln S_t) = \frac{1}{S_t}(rS_t dt + \sigma S_t dW_t^\star) - \frac{1}{2S_t^2}(\sigma^2 S_t^2 dt).$$

Simplifying:

$$d(\ln S_t) = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t^\star. \tag{B.3}$$

Integration

Integrating (A.3) from 0 to T:

$$\int_0^T d(\ln S_t) = \int_0^T \left(r - \frac{1}{2}\sigma^2\right) dt + \int_0^T \sigma dW_t^\star.$$

This yields:

$$\ln S_T - \ln S_0 = \left(r - \frac{1}{2}\sigma^2\right) T + \sigma W_T^\star. \tag{B.4}$$

Normality of the Log-Return

Since $W_T^\star \sim \mathcal{N}(0, T)$, we have:

$$\ln S_T - \ln S_0 \sim \mathcal{N}\left(\left(r - \frac{1}{2}\sigma^2\right) T, \sigma^2 T\right).$$

Therefore:

$$\ln S_T \sim \mathcal{N}(\mu, \text{var}),$$

where:

$$\mu = \ln S_0 + \left(r - \frac{1}{2}\sigma^2\right) T, \qquad \text{var} = \sigma^2 T. \tag{B.5}$$

Lognormal Distribution

If $\ln S_T \sim \mathcal{N}(\mu, \text{var})$, then $S_T = \exp(\ln S_T)$ follows a lognormal distribution with density:

$$f_{S_T}(s) = \frac{1}{s\sqrt{2\pi \text{var}}} \exp\left(-\frac{(\ln s - \mu)^2}{2\text{var}}\right), \quad s > 0. \tag{B.6}$$

Equivalent Representation

Alternatively, we can write S_T as:

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right) T + \sigma\sqrt{T} Z\right), \quad Z \sim \mathcal{N}(0, 1). \tag{B.7}$$

This confirms that S_T is lognormally distributed under the risk-neutral measure.

C Delta Behavior

Delta as a Function of Stock Price S_0

Table 5: Delta as a function of Stock Price S_0 ($K = 30$, $r = 0.04$, $\sigma = 0.15$, $T = 1$)

S_0	10	15	20	25	28	**30**	32	35	40	45	50	60
Δ^a	-1.000	-1.000	-0.982	-0.618	-0.094	**0.267**	0.560	0.829	0.976	0.998	1.000	1.000
Δ^q	-38.710	-28.065	-17.420	-6.775	-0.388	**3.870**	8.128	14.515	25.160	35.805	46.449	67.739

Delta as a Function of Strike Price K

Table 6: Delta as a function of Strike Price K ($S_0 = 30$, $r = 0.04$, $\sigma = 0.15$, $T = 1$)

K	10	15	20	25	28	**30**	32	35	40	45	50
Δ^a	1.000	1.000	0.998	0.881	0.577	**0.267**	-0.071	-0.507	-0.885	-0.982	-0.998
Δ^q	43.870	33.870	23.870	13.870	7.870	**3.870**	-0.130	-6.130	-16.130	-26.130	-36.130

Delta as a Function of Volatility σ

Table 7: Delta as a function of Volatility σ ($S_0 = 30$, $K = 30$, $r = 0.04$, $T = 1$)

σ	0.01	0.02	0.05	0.08	0.12	**0.15**	0.18	0.22	0.30	0.40	0.60	0.80
Δ^a	1.000	0.956	0.591	0.411	0.306	**0.267**	0.245	0.230	0.223	0.236	0.286	0.347
Δ^q	2.455	2.474	2.605	2.850	3.354	**3.870**	4.505	5.546	8.330	13.284	29.510	58.433

Delta as a Function of Time to Maturity T

Table 8: Delta as a function of Time to Maturity T ($S_0 = 30$, $K = 30$, $r = 0.04$, $\sigma = 0.15$)

T	0.05	0.10	0.20	0.40	0.60	0.80	**1.00**	1.25	1.50	2.00	2.50	3.00
Δ^a	0.061	0.086	0.121	0.171	0.209	0.240	**0.267**	0.298	0.324	0.371	0.411	0.446
Δ^q	0.188	0.376	0.755	1.519	2.293	3.076	**3.870**	4.876	5.897	7.989	10.147	12.374

Delta as a Function of Time t

Table 9: Delta as a function of Time t ($S_0 = 30$, $K = 30$, $r = 0.04$, $\sigma = 0.15$, $T = 1$, $S_t = S_0$)

t	**0.00**	0.10	0.20	0.40	0.50	0.60	0.75	0.85	0.90	0.95	0.98	0.99
Δ_t^a	**0.267**	0.254	0.240	0.209	0.191	0.171	0.136	0.105	0.086	0.061	0.039	0.027
Δ_t^q	**3.870**	3.472	3.076	2.293	1.905	1.519	0.945	0.565	0.376	0.188	0.075	0.038

Delta as a Function of Risk-Free Rate r

Table 10: Delta as a function of Risk-Free Rate r ($S_0 = 30$, $K = 30$, $\sigma = 0.15$, $T = 1$)

r	0.000	0.005	0.010	0.015	0.020	**0.040**	0.060	0.080	0.100	0.120
Δ^a	0.060	0.086	0.113	0.139	0.165	**0.267**	0.365	0.457	0.542	0.618
Δ^q	1.365	1.673	1.982	2.293	2.605	**3.870**	5.160	6.476	7.819	9.189

D Product Comparison through Empirical Hedging Costs

A final practical question is whether both products remain equally attractive from the issuer's perspective once realistic hedging costs are taken into account. Although the quadratic payoff may initially appear more profitable due to its higher premium, its larger convexity also generates substantially higher transaction costs throughout the life of the contract.

To quantify this effect, we simulate a hedging portfolio with an initial capital of \$1,000,000 using the Monte Carlo framework developed previously. For each product, we estimate the expected cumulative transaction costs under the different hedging strategies and determine the maximum number of contracts that can be issued while remaining fully hedged. Denoting by V_0 the initial product price, $\overline{TC}$ the average transaction cost per contract, and N the number of contracts sold, the budget constraint becomes

$$N\left(V_0 + \overline{TC}\right) = 1,000,000.$$

This yields

$$N = \frac{1,000,000}{V_0 + \overline{TC}}.$$

Finally, we compare the total expected profit, the capital efficiency and the risk-adjusted profitability of both products. This analysis allows us to determine whether the additional premium received for the quadratic payoff is sufficient to compensate for its significantly larger hedging costs.

Results

Table 11: Comparison of the two products for a $1 000 000 investment portfolio.

	Absolute	Quadratic
Initial Price	3.64	22.76
Mean Transaction Cost	0.17	10.61
Contracts Sold	262 128	29 975
Collected Premium ($)	954 367	682 090
Total Transaction Cost ($)	45 633	317 910
Transaction Cost / Premium	4.78%	46.61%

Although the quadratic payoff has a higher initial value (22.76 versus 3.64), its replication is significantly more expensive. For a portfolio of $1,000,000, approximately 29 975 quadratic contracts can be issued compared with 262 128 absolute contracts. More importantly, total transaction costs represent 46.61% of the collected premium for the quadratic payoff, compared with only 4.78% for the absolute payoff. Consequently, despite generating a larger premium per contract, the quadratic payoff is considerably less attractive from the issuer and buyer's perspectives due to its substantially higher hedging costs.

E Table - Extension 2

Table 12: Summary of the numerical results obtained for the different hedging strategies.

Product	Strategy	Frequency	Mean	Std	RMSE	Avg. TC	Avg. Trades
Absolute	Delta	Daily	-1.5175	0.6179	1.6384	1.4850	251
	Delta	Weekly	-0.6988	0.5123	0.8665	0.6883	50
	Delta	Monthly	-0.3566	0.8724	0.9424	0.3413	11
	Delta–Gamma	Daily	-0.1812	0.0000	0.1812	0.1741	251
	Delta–Gamma	Weekly	-0.1812	0.0000	0.1812	0.1741	50
	Delta–Gamma	Monthly	-0.1812	0.0000	0.1812	0.1741	11
Quadratic	Delta	Daily	-19.3917	4.4073	19.8863	18.9800	251
	Delta	Weekly	-9.0509	5.7220	10.7080	8.7867	50
	Delta	Monthly	-4.4964	10.1589	11.1095	4.2978	11
	Delta–Gamma	Daily	-21.6312	10.2151	23.9219	21.2361	251
	Delta–Gamma	Weekly	-10.8809	5.3176	12.1108	10.6059	50
	Delta–Gamma	Monthly	-5.8776	5.3261	7.9318	5.7102	11
Adaptive	Δ (1σ)	Quadratic	-17.5343	5.7492	18.4528	17.0150	226.07
	Δ (2σ)	Quadratic	-15.2143	8.9390	17.6459	14.7772	194.41
	Δ (3σ)	Quadratic	-12.4307	12.2874	17.4787	12.3617	159.55
	$\Delta\Gamma$ (1σ)	Quadratic	-13.5984	6.6322	15.1295	13.2040	189.03
	$\Delta\Gamma$ (2σ)	Quadratic	-5.6095	9.0747	10.6685	5.4086	34.91
	$\Delta\Gamma$ (3σ)	Quadratic	-2.8799	13.8022	14.0994	2.9885	7.13
	Δ (1σ)	Absolute	-1.2313	0.7262	1.4295	1.2083	204.60
	Δ (2σ)	Absolute	-0.9128	1.1067	1.4346	0.8950	136.64
	Δ (3σ)	Absolute	-0.6404	1.5877	1.7120	0.5905	81.80
	$\Delta\Gamma$ (1σ)	Absolute	-0.1812	0.0000	0.1812	0.1741	251
	$\Delta\Gamma$ (2σ)	Absolute	-0.1812	0.0000	0.1812	0.1741	251
	$\Delta\Gamma$ (3σ)	Absolute	-0.1812	0.0000	0.1812	0.1741	251